

A. Project Title

Home Guardian: Smart Emergency Monitoring Mobile App

B. Background

The inspiration for this project stems from a desire to protect and monitor my own family members, especially those who may be vulnerable or living alone. Many commercial home security and monitoring solutions are prohibitively expensive, require ongoing subscriptions, or lock users into closed ecosystems. Furthermore, these solutions often handle privacy and data externally, requiring users to trust third-party providers with sensitive home audio and video feeds.

My main purpose is to create an affordable, self-hosted, and privacy-centric system that empowers users to monitor their loved ones with confidence and full control. By leveraging off-the-shelf hardware and open-source software, this system can be built and maintained by anyone, without recurring fees. Tailscale integration makes secure remote access simple, overcoming the hurdles of traditional port forwarding and enhancing usability for real families.

C. Objectives

- To empower families with real-time monitoring and instant emergency notifications.
- To enable remote manual control and live viewing of a smart camera system.
- To demonstrate reliable communication between a mobile app and IoT hardware.
- To showcase how smart automation (patrol/face detection, wake word detection) can enhance home safety.
- To provide a self-hosted, privacy-focused, and affordable alternative to commercial solutions.

D. Key Functional Features

1. User Authentication

- Secure login, registration, and logout with Firebase Authentication.

2. Live Camera Feed

- Watch low-latency video stream from the smart camera directly in the app.

3. Manual Camera Control

- Remotely pan and tilt the camera using on-screen slider controls.

4. Patrol Mode with Face Detection

- Start patrol mode: camera sweeps area and stops when a face is detected

5. Emergency Wake Word Detection & Notification

- When a wake word (e.g., “Home Guardian”) is detected via the microphone, the app receives an instant push notification.

6. Event Log / History

- View a chronological list of important events (power on/off, patrol mode start/stop, manual interventions).

E. Data Source & Data Storage

Data Source:

- Live JSON data from Linux system (via REST API and WebSocket), including camera status, face/wake word events, and video stream URL.
- Video stream uses [MediaMTX](#) WebRTC protocol for lower latency.

Data Storage:

- Event logs in Firebase Firestore.
- Local storage (e.g., HiveBox) for adding cameras.

IoT/Backend Technology:

- Ubuntu/Raspberry Pi, USB webcam, built-in microphone, custom pan-tilt kit bracket with two 180-degree servos.
- ESP32 Microcontroller for servo controls.
- Python (Serial connections to ESP32, OpenCV for face detection, Picovoice’s Porcupine wake word detection engine, FastAPI for REST API and WebSockets).
- Firebase Admin SDK for Cloud Messaging (Push notifications)
- Communication via WiFi network and secure device access using Tailscale.

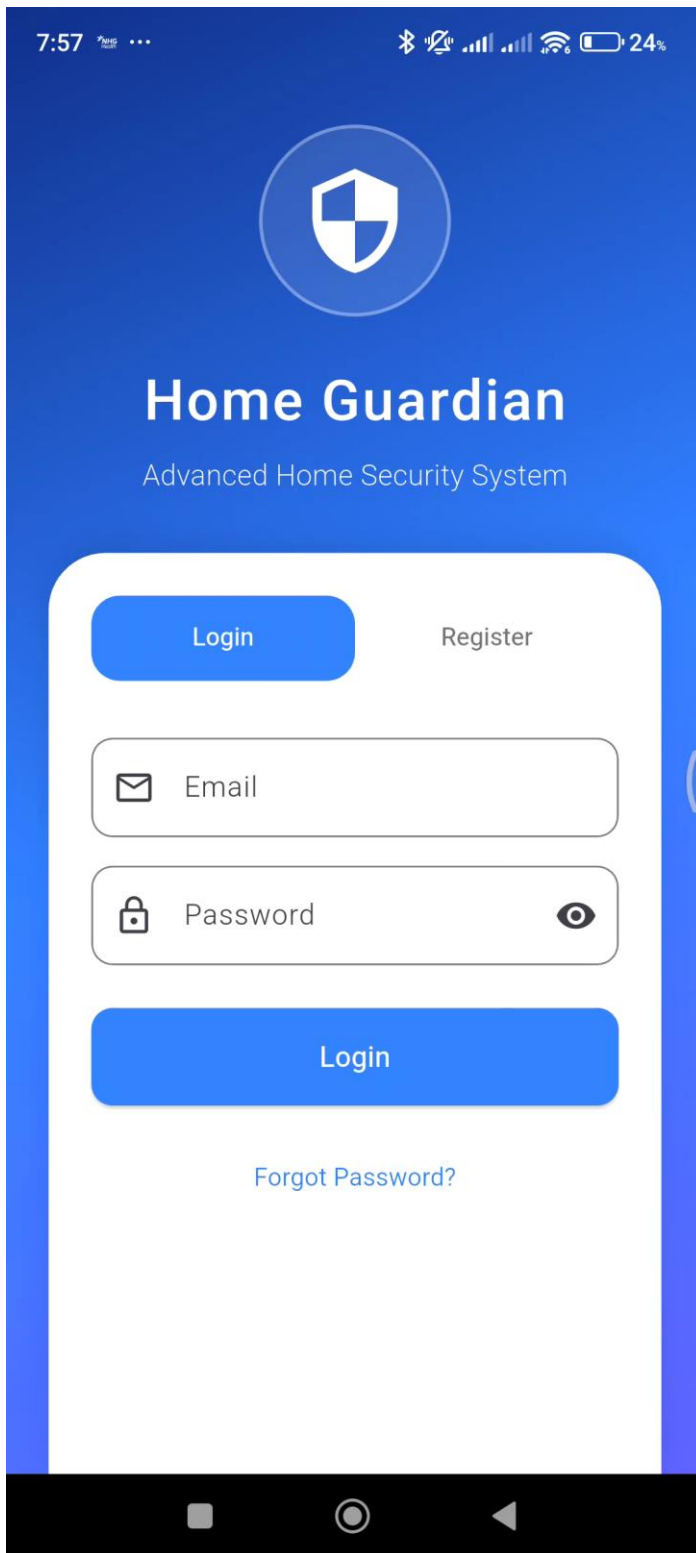
F. State Management

The app will use the Provider package for state management to ensure scalability and maintainability.

G. References

- [Flutter Documentation](#)
- [Firebase Documentation](#)
- [Provider State Management](#)
- [OpenCV Face Detection](#)
- [Porcupine Wake Word Engine](#)
- [FastAPI](#)
- [Tailscale Documentation](#)
- [MediaMTX](#) and [aiortc](#)
- ESP32 [Pinout Reference](#) and Arduino [ESP32Servo Library](#)

H. Screenshots and Pictures



7:55

Bluetooth, Wi-Fi, Cellular, and Battery status icons

Home Guardian



Welcome back,
iamhx



All cameras online and operational

My Cameras

1 camera

HX

Online



hx-ubuntu.emu-snake....
Last checked: 1m ago



Add Camera

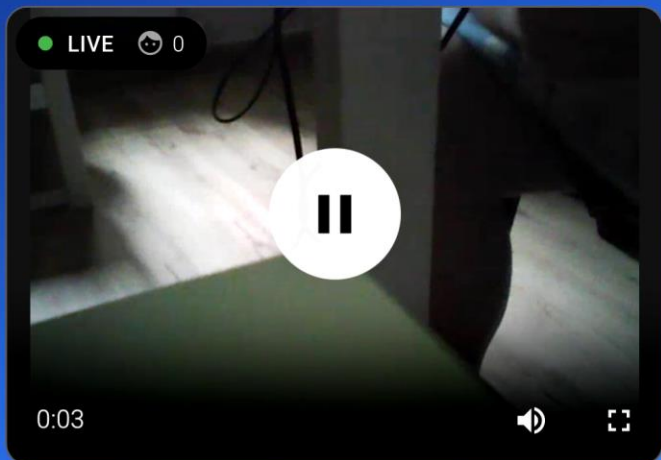
Connect a new Home
Guardian server



7:55

Bluetooth, Signal, Wi-Fi, 24%

< HX



Servos
Inactive

Mode
INACTIVE

 Power On


Patrol


Smart Patrol

 Manual Control

7:55

Bluetooth, Signal, Wi-Fi, 24%

← HX



Servos
Active

Mode
MANUAL

⚠ Power Off

Manual Control



↺ Pan  104° ↻

∨ Tilt  74° ^

 Center

7:57

Bluetooth, Signal, Wi-Fi, 24%

← HX

LIVE 0



Wake Word Detected!

A wake word has been detected by your Home Guardian system.



Detected: 2025-08-13 07:57:10



Status: ACTIVE

OK

Stop Patrol

Smart Patrol



Manual Control

7:57

Bluetooth, Signal, Wi-Fi, 24%

Action History



Patrol started

Camera: HX

Aug 13 • 07:57 AM



Camera powered on

Camera: HX

Aug 13 • 07:56 AM



Camera powered off

Camera: HX

Aug 13 • 07:56 AM



Smart patrol stopped

Camera: HX

Aug 13 • 07:56 AM



Smart patrol started

Camera: HX

Aug 13 • 07:55 AM



Manual control stopped

Camera: HX

Aug 13 • 07:55 AM



Manual control started

Camera: HX

Aug 13 • 07:55 AM



Camera powered on

Camera: HX

Aug 13 • 07:55 AM



Camera powered off

Camera: HX

Aug 13 • 07:55 AM



Manual control stopped

Camera: HX

Aug 13 • 07:55 AM

7:57

Bluetooth, Wi-Fi, Cellular, and Battery status icons

Settings



iamhx

choohongxuan@gmail.com



Settings



About

App information and version



Logout

Sign out of your account



